A logarithmic-derivative comparison for Dirichlet Newman families

Drafted for Edward Baker

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Research draft. Proofs and scalar arithmetic have internal same-model LLM checks; independent specialist review and a priority audit remain outstanding.

Abstract

We give an explicit lower bound for the off-zero logarithmic derivative of a primitive Dirichlet heat deformation. A scalar comparison then controls the time at which all its zeros become real. A real sine tail added to canonical-product truncations preserves the required initial lower bound on the whole real line and allows the comparison to pass to the original function without a compact-confinement hypothesis. In the normalization $s = (1 + iz)/2$, the argument gives $\Lambda_\chi \le 27/100$ for every primitive character of conductor $q \ge 10^{12}$, and $\Lambda_\chi = O((\log q)^{-4/7})$ uniformly as $q \to \infty$. The finite scalar inequalities use exact rational enclosures. Extended proofs and the certificate specification are retained in the accompanying research notes.

1 Normalization and statements

Let χ be primitive of conductor $q > 1$, with $\chi(-1) = (-1)^{\mathfrak{a}}$, $\mathfrak{a} \in \{0, 1\}$. Choose a unit phase α_χ with $\alpha_\chi^2 \varepsilon(\chi) = 1$, where $\varepsilon(\chi) = \tau(\chi)/(i^{\mathfrak{a}}\sqrt{q})$, and set

$$\xi_\chi(s) = \alpha_\chi (q/\pi)^{s/2} \Gamma\big((s + \mathfrak{a})/2\big) L(s, \chi), \tag{1}$$

$$\Phi_\chi(u) = 4\alpha_\chi (\pi/q)^{\mathfrak{a}/2} e^{(1+2\mathfrak{a})u} \sum_{n \ge 1} \chi(n) n^{\mathfrak{a}} e^{-\pi n^2 e^{4u}/q}, \tag{2}$$

$$H_{t,\chi}(z) = \int_{\mathbb{R}} e^{tu^2} \Phi_\chi(u) e^{izu} \, \mathrm{d}u. \tag{3}$$

Then $H_{0,\chi}(z) = \xi_\chi((1 + iz)/2)$ and $\partial_t H_{t,\chi} = -\partial_z^2 H_{t,\chi}$. The Hermitian functional equation makes $H_{t,\chi}$ real entire, although it need not be even. With compatible phases, $H_{t,\bar\chi}(z) = \overline{H_{t,\chi}(-\bar z)}$. The normalization is detailed in Note 1 [5] and agrees with Dobner's framework [1]. Its associated constant Λ_χ is characterized by reality of all zeros exactly for $t \ge \Lambda_\chi$.

Theorem 1.1 (Dirichlet consequence)**.** *Every primitive Dirichlet character of conductor $q \ge 10^{12}$ satisfies*

$$\Lambda_\chi \le \frac{27}{100}.$$

Moreover, uniformly over primitive characters as $q \to \infty$,

$$0 \le \Lambda_\chi = O\big((\log q)^{-4/7}\big), \qquad \Lambda_\chi \longrightarrow 0.$$

These statements use neither GRH, a quasi-RH half-plane, nor an assumed confinement theorem for the character flow.

The lower bound is Dobner's published theorem [1]. The upper bound will follow from a general comparison. For $C > 0$ define

$$D_C(t) = 1 + 2C(1 - e^{-t/2}), \qquad m_C(t) = \frac{Ce^{-t/2}}{D_C(t)}, \tag{4}$$

$$I_C(t) = e^{-t} D_C(t)^{4/3}, \qquad J_C(t) = \int_0^t I_C(s)\,\mathrm{d}s. \tag{5}$$

Theorem 1.2 (Comparison without confinement)**.** *Let F be a nonzero real entire function of order at most one, with all zeros in $|\operatorname{Im} z| \le 1$. Suppose*

$$-\operatorname{Im}\frac{F'}{F}(x + 3i) \ge C > 0 \qquad (x \in \mathbb{R}).$$

If $0 < T \le 1/2$ and $J_C(T) \ge 1/2$, then $e^{-T\partial_z^2} F$ has only real zeros.

The proof first treats flows with confined nonreal zeros, then removes that restriction by a sine-tail approximation. No evenness assumption is present. The extended approximation argument is in Note 3 [7].

2 Canonical sums and a positive-time lower bound

For a real entire function with its zeros $\rho = a + iv$ in a bounded strip, the genus-one product gives, at p above that strip,

$$S(x, p, t) := -\operatorname{Im}\frac{F_t'}{F_t}(x + ip) = \sum_\rho \frac{p - v}{(x - a)^2 + (p - v)^2}. \tag{6}$$

Zeros are counted with multiplicity, including a possible zero at the origin. The imaginary compensators cancel under conjugation. Their absolute summability follows from bounded imaginary parts and $\sum_{\rho \ne 0} |\rho|^{-2} < \infty$. A separate, possibly divergent, real sum of reciprocal zeros is never used.

For the Dirichlet flow, the double-exponential kernel decay gives

$$\log \max_{|z| \le R} |H_{t,\chi}(z)| = O_{\chi,T}((1 + R)\log(2 + R)) \quad (0 \le t \le T). \tag{7}$$

The approximants in Section 4 have exponential-type growth. These bounds also justify the canonical differentiations and completeness of the real characteristics below. Details, including the zero-free anchor and uniform Jensen estimate, appear in Note 2, Section 2 [6].

Proposition 2.1 (Burgers lower bound)**.** *For the Dirichlet flow or the exponential-type approximants below, suppose the zeros remain in $|\operatorname{Im} z| \le 1$ and $S(x, p, 0) \ge C > 0$. Put $d = p - 1 > 0$. Then*

$$S(x, p, t) \ge \frac{Ce^{-2t/d^2}}{1 + dC(1 - e^{-2t/d^2})} \qquad (x \in \mathbb{R},\ 0 \le t \le 1/2). \tag{8}$$

Proof. Write $U = F_t'/F_t = u - iS$ on the line of height p. Equation (6) and its differentiated sums give

$$|u_x| \le S/d, \qquad |S_{xx}| \le 2S/d^2.$$

The heat equation implies $U_t = -U_{xx} - 2UU_x$, hence

$$S_t + 2uS_x = -S_{xx} - 2Su_x.$$

Along $x' = 2u$, this yields $S' \geq -2S/d^2 - 2S^2/d$. The Riccati solution with initial value C is the right side of (8). The characteristics are complete in both time directions: growth and a uniformly zero-free anchor ip imply $|u(x,p,t)| = O((1+|x|)\log(2+|x|))$ on bounded time intervals, which prevents finite-time escape. Their flow is onto $\mathbb{R}$, proving uniformity in x. □

At $p = 3$, the right side is $m_C(t)$ in (4). This is a quantified loss, rather than an assertion that the time-zero lower bound persists unchanged.

3 The comparison when contacts are compact

We briefly retain the confinement hypothesis used in Note 2: on each $[\delta, T]$, $\delta > 0$, the nonreal zeros lie in a bounded horizontal region. This hypothesis will be established for approximants only.

At a highest simple zero $x + ih$, put $Q = h^2$. The conjugate-pair kernel

$$K_h(d,v) = \frac{h-v}{d^2+(h-v)^2} + \frac{h+v}{d^2+(h+v)^2}$$

obeys $K_h/h \geq K_p/p$ whenever $|v| \leq h$ and $p^2 \geq 5h^2$. Clearing denominators reduces this to nonnegativity of

$$D^2 + (A+B-6r)D + (A-r)(B-r), \quad D = d^2,\ A = h^2,\ B = p^2,\ 0 \leq r = v^2 \leq A.$$

The zero-motion identity, with the zero's own conjugate pair removed, therefore gives

$$Q' \leq -2 - \frac{4Q}{p} S(x,p,t) + \frac{8Q}{p^2 - Q}. \tag{9}$$

This form of the comparison appears in Hakimoglu-Brown [4]; the algebra and its non-even extension are recorded in Notes 1–2.

Classical strip contraction gives $Q \leq 1$. At $p = 3$, $8/(9-Q) \leq 1$, so Proposition 2.1 gives

$$Q' \leq -2 - \left(\frac{4}{3} m_C(t) - 1\right) Q.$$

The corresponding solution from one is

$$\overline{Q}(t) = \frac{1 - 2J_C(t)}{I_C(t)}. \tag{10}$$

Until it reaches zero it decreases, with descent rate at least $2 - \overline{Q} \geq 1$.

For completeness, use a strictly higher comparison curve, with initial value $1 + \varepsilon$ and a positive ε added to its differential equation. This supplies an initial time gap uniformly in space. Confinement makes a later first contact compact. A simple contact contradicts (9); a multiple contact has a backward Hermite split above the smooth curve. Letting $\varepsilon \downarrow 0$ yields $M(t) \leq \overline{Q}(t)$ while the latter is positive. If it reaches zero at T_0, strip contraction makes $F_{t+\overline{Q}(t)/2}$ real-rooted for $t < T_0$. Heat continuity and Hurwitz then make F_{T_0} real-rooted; strip contraction preserves that property afterwards. The extended contact proof is Note 2, Section 4 [6].

4 Removing confinement by a real sine tail

Fix $0 < C_1 < C$, where C is the input in Theorem 1.2, and write

$$F(z) = ce^{az}z^m \prod_{\rho\neq 0} E_1(z/\rho), \qquad E_1(w) = (1-w)e^w, \quad c, a \in \mathbb{R}.$$

Let P_R truncate this product at $|\rho| \le R$, retaining conjugate pairs and multiplicities. Set

$$\kappa = 2(C_1+1), \quad h = \pi/\kappa, \quad M = \lfloor R/(4h) \rfloor, \qquad G_R(z) = P_R(z) \prod_{k>M} \left(1 - \frac{z^2}{k^2h^2}\right). \tag{11}$$

The new zeros are real, and $G_R \to F$ locally uniformly.

Lemma 4.1. *For all sufficiently large R, $S_{G_R}(x, 3, 0) \ge C_1$ for every real x. Each fixed approximant flow has uniformly confined nonreal zeros on every bounded time interval. Moreover, $(G_R)_t \to F_t$ locally uniformly, uniformly on bounded nonnegative time intervals.*

Proof. We give the estimates that carry the argument; their extended proofs are in Note 3, Sections 2–4 [7]. For $|x| \le R/2$, the omitted true-zero Poisson tail is at most

$$16(p+1) \sum_{|\rho|>R} |\rho|^{-2} = o(1).$$

For $|x| \ge R/2$, the full real lattice has lower bound $\kappa \tanh(\kappa p)$, and removing its central zeros costs at most $8p/(hR) + 16p/R^2$. At $p = 3$ this leaves a bound exceeding C_1; the strict input gap $C - C_1$ handles the central region.

For fixed R, the sine product rewrites G_R as $e^{b_R z} r_R(z) \sin(\kappa z)$ with b_R real and r_R real rational, its poles canceled by sine zeros. If $r_R(z) \sim c_R z^{d_R}$ at infinity, Gaussian imaginary translation gives, uniformly for $|\operatorname{Im} z| \le 1$ and bounded t,

$$\frac{(G_R)_t(z)}{c_R z^{d_R} e^{b_R z + (\kappa^2 - b_R^2)t}} = \sin\bigl(\kappa(z - 2b_R t)\bigr) + O_{R,T}(|\operatorname{Re} z|^{-1}). \tag{12}$$

Along the Gaussian line the real part remains $\operatorname{Re} z$, so no rational pole is crossed. Rouché disks about the distant real sine zeros contain exactly one zero each; reality forces it to be real. The complement in the strip is zero-free. The strip itself follows from the genus-one operator theorem of Cardon [2, Theorem 1.5], with quadratic parameter $2t$ and no quadratic exponential in G_R.

Finally, for any fixed $1 < \sigma < 2$, canonical-product bounds give $|P_R(z)| \le e^{A(1+|z|)^\sigma}$ uniformly in R. The added tail is bounded by $\sinh(\kappa|z|)/(\kappa|z|) \le e^{\kappa|z|}$. These provide a common sub-Gaussian majorant in

$$F_t(z) = \frac{1}{\sqrt{\pi}} \int_{\mathbb{R}} e^{-v^2} F(z + 2i\sqrt{t}\, v)\, \mathrm{d}v.$$

Dominated convergence proves the heat-limit assertion. The limit is nonzero: real Gaussian translation inverts imaginary Gaussian translation under this same majorant (Note 3, Section 4). □

Apply the compact comparison to G_R. If $J_C(T) > 1/2$, choose $C_1 < C$ close enough that $J_{C_1}(T) > 1/2$. The lemma makes $(G_R)_T$ real-rooted. Hurwitz's theorem in each open half-plane passes this to the nonzero limit F_T. For equality, first use $T + \eta < 1/2$ and then let $\eta \downarrow 0$; the case $T = 1/2$ is classical strip contraction. This proves Theorem 1.2. No bound uniform in R on the approximants' confinement regions is required.

5 Arithmetic input and exact scalar arithmetic

The initial Dirichlet logarithmic derivative obeys

$$S_\chi(x,p,0) = \frac{1}{2}\operatorname{Re}\frac{\xi_\chi'}{\xi_\chi}\left(\frac{1+p+ix}{2}\right), \tag{13}$$

$$S_\chi(x,p,0) \geq C_{q,\mathfrak{a}}(p) := \frac{1}{4}\left[\log\frac{q}{\pi} + \psi\left(\frac{1+p+2\mathfrak{a}}{4}\right)\right] - \frac{1}{2}\left(-\frac{\zeta'}{\zeta}\right)\left(\frac{1+p}{2}\right). \tag{14}$$

Indeed, the Hermitian functional equation gives (13); the gamma logarithmic derivative is bounded below by its value on the real axis, and the Euler product gives $|L'/L(s,\chi)| \leq -\zeta'/\zeta(\operatorname{Re} s)$ for $\operatorname{Re} s > 1$. At $p = 3$ the even case is the smaller of the two parities.

The finite certificate in Note 2, Section 5 [6], proves

$$C_{q,\mathfrak{a}}(3) > \frac{1221}{200} = 6.105 > 6 \qquad (q \geq 10^{12},\ \mathfrak{a} = 0,1). \tag{15}$$

It uses $\log 10 > 23/10$, $\log 4 < 7/5$, $\pi < 4$, $\gamma < H_{1000} - \log 1000 < 29/50$, and a prime-power sum through 1000 with the positive tail $(\log 1000 + 1)/1000$ to prove $(-\zeta'/\zeta)(2) < 3/5$. All decisions use exact rational enclosures.

The integral in (5) has the elementary form

$$J_C(t) = \frac{3(1+2C)}{14C^2}(D_C(t)^{7/3} - 1) - \frac{3}{20C^2}(D_C(t)^{10/3} - 1). \tag{16}$$

For $C = 6$, $T = 27/100$, the same certificate encloses

$$0.502464325582617834 \leq J_6(T) \leq 0.502464325582617835.$$

Alternating exponential series and rational cube-root bisection make the lower endpoint rigorous. Theorem 1.2 and (15) prove the explicit part of Theorem 1.1. The certificate programme, parameters, hashes, and extended scalar verification are referenced from Notes 2–3 rather than reproduced here. Its original record was conditional on an analytic contract; Section 4 supplies that contract without changing the arithmetic.

For $0 \leq t \leq 1/2$,

$$J_C(t) \geq e^{-1/2}\frac{(1 + Ce^{-1/4}t)^{7/3} - 1}{(7/3)Ce^{-1/4}}.$$

Consequently

$$B(C) := \frac{e^{1/4}}{C}\left[\left(1 + \frac{7}{6}e^{1/4}C\right)^{3/7} - 1\right] \tag{17}$$

is a sufficient reality time whenever $B(C) \leq 1/2$ and $C \leq C_{q,\mathfrak{a}}(3)$. Since $C_{q,\mathfrak{a}}(3) = \frac{1}{4}\log q + O(1)$ uniformly in parity, $B(C) = O((\log q)^{-4/7})$ for the natural input. This proves the remaining assertions of Theorem 1.1.

6 Scope, controls, and provenance

The sine-tail limit bypasses confinement for the original Dirichlet flow; it does not prove that separate assertion. A periodic real-entire heat flow may have a positive initial logarithmic-derivative

floor and unboundedly many nonreal zeros at positive times. The example and its scope are in Note 2, Section 7. This does not contradict the present quantitative comparison.

The arithmetic gain enters through (14). Euler-factor collision controls can violate its precise Euler-product allowance. Weighted-square, reflected, and primitive-compatible controls need not have the character completion or that allowance. The general comparison remains valid when their actual input floor meets its assumptions; no character-specific lower bound is assigned to them by analogy. This is smoothing at positive time, not a proof of GRH at time zero.

A literature correction is necessary. Ki–Kim–Lee [3, Section 6, Example 1] already extend their qualitative strict bound below 1/2 to real primitive Dirichlet characters. The initial investigation's search missed that precedent. Their example for a complex character uses a real self-dual combination, rather than the individual non-even family in (3). We make no priority claim for the quantitative results of this draft.

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References

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[5] Research Note 1, prepared for E. Baker with Claude Opus 5.5, *Normalization and transfer contract*, 17 October 2026, notes/round-1.

[6] Research Note 2, prepared for E. Baker with GPT-6.1-sol, *An explicit positive-time floor and a comparison conditional on confinement*, 17 October 2026, notes/round-2.

[7] Research Note 3, prepared for E. Baker with GPT-6.1-sol, *Removing confinement from the strip comparison by a real sine tail*, 17 October 2026, notes/round-3.